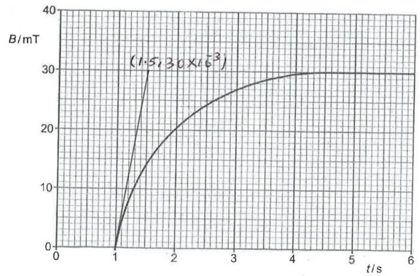
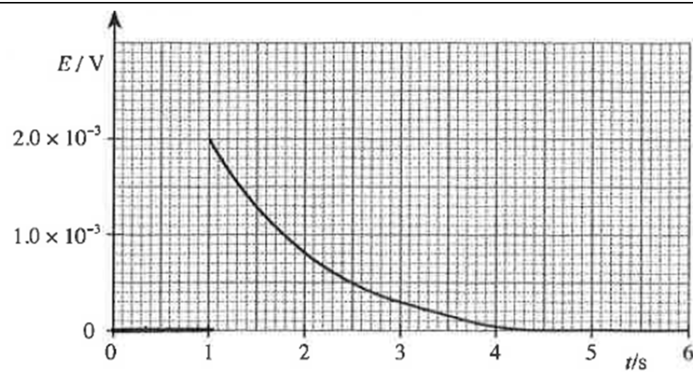
<table>
<tr><td></td><td></td></tr>
<tr><td>6(a)</td><td>Faraday’s law of electromagnetic induction states that the induced e.m.f. in a coil is directly proportional to the rate of change of magnetic flux linkage through the coil.</td></tr>
<tr><td>(b)(i)</td><td>

Gradient of the tangent at time t =1.0 s $=\dfrac{30\times10^{-3}-0}{1.5-1.0}=6.0\times10^{-2}\ \text{T s}^{-1}$</td></tr>
<tr><td>(b)(ii)</td><td>Induced e.m.f $=\dfrac{\text{d}\left(NBA\cos\theta\right)}{\text{d}t}=NA\dfrac{\text{d}B}{\text{d}t}=140\times2.4\times10^{-4}\times6.0\times10^{-2}$
$= 2.0\times10^{-3}$ V</td></tr>
<tr><td>(b)(iii)</td><td>
</td></tr>
<tr><td>7(a)(i)</td><td>Binding energy of a nucleus is the minimum energy required to break the nucleus</td></tr>
</table>